Sanvi Nowal

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.46
Shalom Hills International School	Class XII (CBSE)	2023	97.00%
Shalom Hills International School	Class X (CBSE)	2021	99.00%

SCHOLASTIC ACHIEVEMENTS & WORK EXPERIENCE

1. Technical Intern at Sprih (an AI-native SaaS based company) for the duration of Jan 2026 to July 2026.
2. Research Intern at Jawaharlal Nehru Centre for Advanced Scientific Research (JNCASR) for the duration of May 2025 to July 2025.
3. Recipient of IITH's Academic Excellence Award for year 2025-2026.

SKILLS

1. **Language Proficiency:** C, C++, Python (with extensive experience in libraries such as NumPy, pandas, scikit-learn, matplotlib, seaborn, SciPy, Tensorflow, PyTorch, CUDA), Julia, SQL, MatLab, NodeJS, CSS.
2. **Machine Learning**: Supervised and Unsupervised Learning, Classical ML Algorithms, Deep Learning, Computer Vision, Ensemble Learning, Model Validation and Evaluation.
3. **LLM & Agents**: Agentic Systems, Multi-Agent Orchestration, LangChain Ecosystem, Claude API, Context Management, Knowledge Graphs, Agent Safety and Sandboxing.
4. **Frontend & Full Stack:** Responsive UI/UX Design, Office Add-In Development, GDrive Connection Handling, Client-Side State and Token Handling, End-to-End Product Delivery.
5. **Backend & Cloud**: Authentication Handling (JWT, Token-based, OAuth), Multi-Tenant Design, REST APIs, Streaming APIs, Relational and Graph Databases (MariaDB, Neo4j), Cloud Deployment (GCP, AWS), Containerization (Docker), Reverse Proxying and TLS with Nginx.
6. **Scientific Computing and Tools**: Streamlit, Aspen Plus, LAMMPS Simulation Software, OVITO, RoboFlow, Density Functional Theory (DFT), Cantera.

PROJECTS

SustainSense: Sprih's Companion AI Product connected to pre-existing data driven pipeline

1. Production-ready and already shipped as the base code of Sprih's second product: Deployed on GCP**,** adopted by the internal teams, demoed to both existing and potential clients.
2. Architected a LangGraph multi-agent system automating a range of applications such as carbon credit purchasing, ESG-linked financing, questionnaire filling and climate risk analysis.
3. Delivered the UI/UX as an MS Word Add-In (Office.js) with inline edit approve/reject controls and built-in Google Drive sync.
4. Engineered a sandboxed terminal over S3-backed thread-scoped filesystems with SHA256 state fingerprinting for agent terminal access with extensive safeguards.

Kinetic-based regularization method for data science applications

1. Reframed the interpolation problem as an energy minimization problem of a physical system, mapping kernel spread to temperature and validation error to an energy landscape, where over-fitting and under-fitting become "cold death" and "hot death".
2. Formulated a kinetic-theory regularization that constrains low order moments of a distribution, cutting interpolation RMSE by upto 16x in 1D and 3x on 2D Franke.
3. Designed an EM-style iterative search for kernel spread with decaying relaxation and coarse-to-fine early stopping.
4. Benchmarked against RBF, Additive GPR, and cubic interpolation: outperformed RBF on random sampling, matched GPR in sparse 6D, and beat cubic on noisy Copernicus climate data (R^2 0.985 vs 0.965).

Instance Segmentation on Experimentally Obtained Calcium Cell Dataset

1. Achieved 0.537 mask AP and 0.773 AP50 on experimentally acquired calcium cell microscopy images by fine-tuning Detectron2 Mask R-CNN (ResNet50-FPN) from COCO-pretrained weights on a 32-image dataset.
2. Built the data pipeline with Roboflow polygon annotations exported to COCO JSON, applying flip and Gaussian noise augmentation to expand the training set under severe data scarcity.

3. Validated robustness with 5-fold cross-validation on a held-out test set, retaining ~90% of performance with only 50% of training data.

Mini Projects completed during Epoch tenure:

1. **Recommender System**: Implemented Bagging, AdaBoost, and stacking from scratch on decision trees to power a personalized song rating predictor on the Million Song Dataset.
2. **Species Evolution Tracker:** Built agglomerative clustering from scratch and extended it beyond a single snapshot to track cluster evolution across 20 time steps of a 1,000-species ecosystem, handling new species and mutations with dendrograms.
3. Implementing SVD from scratch and applying low-rank approximation to compress artwork images, analyzing the compression vs quality trade-off across k via Streamlit GUI.
4. Implementing a computational graph from scratch to simulate back-propagation in a neural network, for integration in AutoGrad system.

Core Projects undertaken:

1. **Catalyst Modeling:** Running a DFT Study of CVD-synthesized Ni/NiO/N-doped graphene electrocatalyst on Urea Electrolysis.
2. **Solving Stiff ODEs:** Application of a Modified Projective Euler scheme as an integrator, that allows larger explicit jumps on stiff kinetics with no Jacobian to simulate Methane Combustion reaction.

RELEVANT COURSES

1. **LangChain Courses:** LangChain Essentials, LangGraph Essentials, LangSmith Essentials, Deep Research with LangGraph.
2. **Neo4j Courses:** Neo4j Fundamentals, Cypher Fundamentals, Graph Data Modeling Fundamentals, Constructing Knowledge Graphs with Neo4j GraphRAG for Python.
3. **Coursera Courses**: Supervised Machine Learning: Regression and Classification (DeepLearningAI x Stanford)
4. **College Introductory Technical Courses**: Introduction to Programming, Introduction to Artificial Intelligence
5. **College Advanced Technical Courses:** Data Structures & Applications, Accelerating Molecular Dynamics with AI/ML, Computer vision in Chemical and Life Science, AI/ML Tool Chains for Chemical Engineers
6. **College Foundational and Advanced Math Courses:** Calculus I and II, Linear Algebra, Differential Equations, Numerical Methods, Advanced Numerical Methods.
7. **Soft Skill Courses**: Communication Skills, Spoken Japanese Basics, Psychology of Interpersonal Relationships

POSITIONS OF RESPONSIBILITY

1. **Core** at Epoch (the AI/ML club of IIT Hyderabad) for Year 2024-2025.
2. **Class Representative** for Chemical Engineering Branch for Year 2023-24.
3. **Student DUGC** (Department Under-Graduate Committee) for Year 2025-2026.
4. **Cup Captain** for **InterIIT Culturals 8.0** Contingent for Literary Arts Cup for Year 2025-2026, and participant for IICM 7.0.
5. **Head of Public Relations and Design** for Behind The Lens (BTL), the filmmaking club at IITH.
6. **Head of Literary Society** (LitSoc) at IITH in 2025-2026, and Core position in the preceding year